Model question paper-(15) 3212

THIRD SEMESTER DIPLOMA EXAMINATION IN INSTRUMENTATION ENGINEERING

DIGITAL CIRCUITS

TIME : 3HRS

MAXIMUM MARKS : 100

PART-A

(Maximum marks:10)

I.Answer the following questions in one or two sentences. Each question carries 2 marks

1.List the various digital codes?

2.Define AC noise margin

3 List the inputs of flip flop

4 Define capacity of memory

5. Define glitch 5x2=10

PART-B

(Maximum marks:30)

II. Answer *any five of* the following questions . Each question carries 6 marks.

1.Convert

i) $(3215.2565)_{10}$ =$(----------)_8$ ii)($3A1BF.51A)_{16}$ =$(\ldots\ldots\ldots\ldots\ldots)_{10}$

iii) $(132157)_8$ =$(------)_{16}$

2.Explain with the help of circuit diagram ,the operation of a TTL inverter

3.Draw the logic diagram of edge triggered JK flip flops and explain

4.Explain the applications of shift registers

5. Draw the logic diagram of Half adder. Give its truth table and realise it

6. Draw the logic diagram of RAM cell and explain its operation

7. Explain the operation of R-2R D/A converter 5x6=30

PART C

(Maximum marks:60)

(Answer one full question from each module .Each full question carries 15 marks)

Module 1

III.(a). Solve the following

i).$(11100.111)_2$= $(-------------\_)_{10}$ ii) using 2's compliment method $(8A)_{16}$-$(16)_{16}$

iii) 110101_2 - 10111_2 using 2's compliment method 2x3=6

(b). Simplify the expression F= $\bar{A}\,\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + AB\bar{C}$ +ABC+$A\bar{B}C$ 9

OR

IV (a).Simplify the Boolean function $F(A,B,C,D)=\sum_m(0,2,3,4,6,9,11,14,15)+\sum_d(1,5,10)$.using K-map 9

(b).Convert

i) 11001111_2 into GRAY code ii) $111011_{(gray)}$ into Binary code

2x3=6

Module2

V. (a).What is full Adder ? Give its truth table and realize it using basic gates 8

(b).Explain the working of 3 bit binary decoder with its logic diagram 7

OR

VI.(a) Explain the working of BCD to decimal decoder. Write its decoding functions and sketch its logic diagram. 8

(b). Explain the working of 4 bit Parallel binary adder 7

Module 3

VII.(a).Explain SR latch with help of diagram and truth table. 7

(b).Design mod 9 synchronous counter. 8

OR

VIII(a). Design mod-7 asynchronous counter. 8

(b). Explain the operation of serial in parallel out shift registers 7

Module4

IX.(a). Explain the working of Single slope ADC 7

(b).Compare Rom and Ram 8

OR

X.(a) Draw and explain the circuit of a typical ROM cell 7

(b). Explain the working of SA A/D Converter 8